

INDIVIDUAL PROJECT LOGBOOK

Project Title: Cloud-Native Learning Management System (LMS) with Layered Architecture

Student Name: Eashenraj A/L Nagarajan

Academic Role: Backend & Deployment Lead

Group Number: Group 4

1. Team Information

Team Members	Role
Eashenraj A/L Nagarajan	Backend & Deployment Lead
Mohammad Adam Basheer B. Mohd Azrul	Frontend & Database Lead
Ammar Haziq B. Saiful Bahri	Full Stack & Architecture Lead

1.1 Project Overview

The Cloud-Native Learning Management System (LMS) is a platform developed to manage educational activities including course creation, student enrollment, content delivery, and assignment submission. The system is built following the Layered Architecture Pattern, separating responsibilities into four distinct layers: Presentation, Business Logic, Data Access, and Database.

The prototype is fully deployed across three cloud platforms:

Frontend (Presentation Layer) — React with Vite, deployed on Netlify

Backend (Business Logic + Data Access Layer) — Spring Boot, deployed on Render

Database Layer — Aiven MySQL cloud-managed relational database

1.2 Individual Role Summary

As the Backend and Deployment Lead, my primary responsibilities throughout the project covered:

Setting up and configuring the Spring Boot backend project with Maven

Implementing the business logic layer including UserService, CourseService, AssignmentService, ContentService, and SubmissionService

Configuring the Aiven MySQL cloud database connection with SSL and deploying to Render

Debugging and resolving backend authentication, CORS, and SSL connection issues

Testing all REST API endpoints using Postman before and after deployment

Contributing to architecture design decisions and the project report

2. Weekly Progress

This section documents the weekly activities, individual contributions, challenges encountered, and action plans undertaken across all project phases.

2.1 Phase 1 — Project Proposal (Week 5)

Week 5 Project Proposal	
Tasks Undertaken	Discussed project idea and scope with the team. Identified the problem statement around traditional monolithic LMS designs. Drafted the project proposal covering system objectives, architecture approach, functional requirements, and non-functional requirements.
Individual Contributions	Contributed to writing the problem description and system objectives sections of the proposal. Helped define the four-layer architecture scope and identified the cloud deployment tools to be used for each layer.
Challenges Faced	The team had difficulty agreeing on the scope of features to include within the project timeline. There was also an initial misunderstanding about how Postman fits into the system architecture.
Action Plan	Held a team discussion to narrow the scope to four core modules: User Management, Course Management, Assignment and Assessment, and Content Delivery. Clarified that Postman is an API testing tool and not part of the deployed system.

2.2 Phase 2 — Architecture Design (Week 7)

Week 7 Architecture Design	
Tasks Undertaken	Designed the full system architecture following the Layered Architecture Pattern. Produced the UML deployment diagram showing the communication flow between Netlify, Render, and Aiven MySQL. Identified the service and component structure across all four layers and documented the Service and Component Identification section.

Individual Contributions	Led the architecture design in the deployment diagram
Challenges Faced	There was confusion about whether the Data Access Layer should be shown as a separate deployment node or nested inside the Render backend container in the UML deployment diagram.
Action Plan	Researched UML deployment diagram conventions and determined that the DAL should be nested inside the Render container since both the business logic and data access layers run within the same Spring Boot JAR file on the same server.

2.3 Project Observation 1 (Week 9)

Project Observation 1 Week 9	
Tasks Undertaken	Began prototype development. Set up the Spring Boot project structure with Maven. Configured the Aiven MySQL cloud database and established the JDBC connection from the backend.
Individual Contributions	Set up the Spring Boot backend project from scratch and created the Maven pom.xml with all required dependencies, including Spring Data JPA, Spring Security Crypto, and the MySQL JDBC connector. Configured application.properties with the Aiven MySQL SSL connection string.
Challenges Faced	The Aiven MySQL connection kept failing on application startup with an SSL handshake error. The error was caused by incorrect SSL parameter syntax in the JDBC connection string.
Action Plan	Identified that the JDBC connector requires sslMode=REQUIRED instead of ssl-mode=REQUIRED. Also added the Hibernate MySQL dialect property to ensure correct SQL generation.

2.4 Phase 3 — Prototype Progress (Week 11)

Week 11 Prototype Progress	
Tasks Undertaken	Continued backend development. Implemented all service classes with full business logic, REST API controllers, and deployed the complete backend to Render.
Individual Contributions	Implemented UserService with BCrypt password hashing and matric number uniqueness validation. Implemented CourseService with three-layer enrollment validation: role check, capacity check, and duplicate enrollment prevention. Deployed the Spring Boot JAR to Render as a web service and configured environment variables.
Challenges Faced	After deployment, registration, and login kept returning a credential error even though the Render build logs showed a successful deployment. The backend was running, but authentication was failing on every request.
Action Plan	Traced the issue through the browser network tab and identified that the frontend registration payload was sending the field name password, but the backend User model expected passwordHash.

2.5 Project Observation 2 (Week 12)

Project Observation 2 Week 12	
Tasks Undertaken	Tested all deployed API endpoints using Postman to verify correct request and response behaviour. Fixed CORS configuration issues between Netlify and Render.
Individual Contributions	identified and resolved a CORS mismatch in WebConfig.java where the allowed origin URL did not exactly match the live Netlify deployment URL. Verified all endpoints returned correct responses in Postman.
Challenges Faced	After enrollment, the student dashboard was not showing the enrolled courses. The enrollment was being saved correctly to the course_enrollments table in Aiven MySQL, but the frontend had no way to retrieve and display only the courses a specific student was enrolled in.
Action Plan	Added a custom JPQL query in CourseRepository using the @Query annotation to find all courses a student is enrolled in.

2.6 Phase 4 — Project Report (Week 13)

Phase 4 Week 13 Project Report & Individual Logbook	
Tasks Undertaken	Compiled the full project report covering architecture design, functional and non-functional requirements, technology stack, service and component identification, system diagrams, team contributions, and conclusion. Compiled the individual logbook.
Individual Contributions	Wrote and reviewed the architecture evaluation, technology stack section, service and component identification sections of the report. Ensured all tool names, endpoint paths, class names, and table names in the report were accurate and consistent with the actual implemented codebase.
Challenges Faced	Ensuring the report accurately reflected the final implemented system rather than the originally planned design.
Action Plan	Cross-referenced every report section against the actual codebase to verify accuracy.

2.7 Phase 5 — Final Demonstration (Week 14)

Phase 5 Week 14 Final Demonstration	
Tasks Undertaken	Prepared the final system demonstration. Ensured all three cloud services (Netlify, Render, Aiven) were running and responsive.
Individual Contributions	Verified all three cloud platforms were active before the demonstration. Prepared verbal explanations of the layered architecture, separation of concerns, and the role of each tool in the deployment stack.
Challenges Faced	Render's free tier Scale-to-Zero feature causes a cold start delay of up to 30 seconds when the backend service has been idle, which could disrupt the demonstration flow.
Action Plan	Opened the live application several minutes before the scheduled demonstration to trigger the Render service to wake up, ensuring instant API responses during the actual demo session.

3. Generative AI Usage Record

All use of Generative AI tools during this project is transparently declared below. All AI-generated outputs were reviewed, verified, and adapted before being applied to any part of the project or report.

Total declared GenAI interactions: 7 entries across Weeks 5 to 13.

3.1 Entry 1 — Week 5

GenAI Entry 1 Week 5 Phase 1: Project Proposal	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	Review the project report introduction and problem description for clarity, flow, and accuracy
Prompt Used	"Read my report and tell me if everything makes sense and is in a good flow"
Output Summary	Identified that Postman was incorrectly described as a REST API rather than a testing tool. Identified repetition in the problem description and suggested a restructured version with cleaner paragraph separation.
How It Was Used	Used the feedback to correct the Postman description and restructure the problem description before Phase 1 submission.

3.2 Entry 2 — Week 7

GenAI Entry 2 Week 7 Phase 2: Architecture Design	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	Generate a UML-compliant deployment diagram for the LMS system architecture
Prompt Used	"Help me generate a UML deployment diagram suitable for this system using React, Spring Boot, and MySQL on Netlify, Render, and Aiven"
Output Summary	Generated a UML deployment diagram with correct stereotypes including executionEnvironment, component, artifact, database, and table. Showed correct nesting of the business logic and data access layers inside the Render backend container.
How It Was Used	Used the diagram as a visual reference for the architecture section. Referring to the generated output, the final diagram was designed using VisualParadigm.

3.3 Entry 3 — Week 7

GenAI Entry 3 Week 7 Phase 2: Architecture Design	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	Identify all system services and components for the Service and Component Identification report section
Prompt Used	"Since you have the whole idea of the system, what are all the service components? I need to do a report part for Service and Component Identification"
Output Summary	Produced a structured breakdown of four modules with business logic service components and DAL DAO components mapped to each module, presented as a descriptive list and summary table.
How It Was Used	Used as the foundation for Section 4.2 Service and Component Identification in the report. The output was reviewed against the actual implemented codebase and trimmed to match only the features that were built.

3.4 Entry 4 — Week 9

GenAI Entry 4 Week 9 Project Observation 1	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	Get clear explanations of all tools and their functions to prepare for the lecturer presentation
Prompt Used	"Explain all the tools involved so I can explain to my lecturer — Axios, Spring Boot, REST API, and more with their functions"
Output Summary	Produced a detailed breakdown of every tool in the system, from React and Axios on the frontend to JDBC and Hibernate on the backend, with a full request-response flow summary table.
How It Was Used	Used to prepare verbal explanations for the Project Observation 1 session with the lecturer and to inform the Technology Stack section of the final report.

3.5 Entry 5 — Week 11

GenAI Entry 5 Week 11 Phase 3: Prototype Progress	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	Diagnose and fix authentication failures on the live deployed system
Prompt Used	"Why failed to register"
Output Summary	Identified four bugs: registration payload used password instead of passwordHash, role enum values sent as Student instead of STUDENT, login redirect checked wrong role casing, and UserService crashed on null email. Produced corrected Login.jsx, UserService.java, User.java, and application.properties files.
How It Was Used	Applied all four corrected files to the project, committed and pushed to GitHub, and redeployed to Render and Netlify. Verified that registration and login worked correctly on the live system after the fix.

3.6 Entry 6 — Week 12

GenAI Entry 6 Week 12 Project Observation 2	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	Improve the student dashboard to store and display enrolled course data per user properly
Prompt Used	"Improve my code where once enrolled in a course it actually stores the course data under that user"
Output Summary	Produced updated CourseRepository.java with a custom JPQL query, CourseService.java with getEnrolledCourses method and duplicate enrollment prevention.
How It Was Used	Applied all four updated files to the project, redeployed the backend to Render and frontend to Netlify via GitHub push, and verified that enrolled courses appeared correctly under the My Courses tab after enrollment.

3.7 Entry 7 — Week 13

GenAI Entry 7 Week 13 Phase 4: Project Report	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	Update all report sections to accurately reflect the final implemented and deployed system
Prompt Used	"I will give you parts of my report, identify changes based on how we developed the project, and whether the right tools were used. Keep my sentences"
Output Summary	Updated report sections to accurately reflect the deployed system with correct tool names, class names, and endpoint paths.
How It Was Used	Reviewed every AI-updated paragraph against the actual codebase before including it in the final report. Sections were further edited manually to match the team writing style and to remove content referencing features not present in the final prototype.

4. Academic Integrity Declaration

I hereby declare that all work documented in this individual logbook is a true and accurate record of my contributions to the Cloud-Native Learning Management System group project. I acknowledge and accept the following:

All tasks, contributions, and activities recorded in Section 2 represent my genuine individual work and involvement in the project.

All use of Generative AI tools has been fully and transparently declared in Section 3, including the tool used, the purpose, the prompt submitted, the output received, and how the output was applied.

I understand that any undeclared use of Generative AI tools or misrepresentation of individual contributions constitutes a breach of the academic integrity policy.

I confirm that the contributions attributed to me in the Team Contributions section of the group report are accurate and reflect my actual role in the project.

Full Name	Eashenraj A/L Nagarajan
Matrix No	S76773
Group Number	Group 4
Academic Role	Backend & Deployment Lead

INDIVIDUAL PROJECT LOGBOOK

Project Title: Cloud-Native Learning Management System (LMS) with Layered Architecture

Student Name: Mohammad Adam Basheer Bin Mohd Azrul

Academic Role: Frontend & Database Lead

Group Number: Group 4

1. Team Information

Team Members	Role
Eashenraj A/L Nagarajan	Backend & Deployment Lead
Mohammad Adam Basheer B. Mohd Azrul	Frontend & Database Lead
Ammar Haziq B. Saiful Bahri	Full Stack & Architecture Lead

1.1 Project Overview

The Cloud-Native Learning Management System (LMS) is a platform developed to manage educational activities including course creation, student enrollment, content delivery, assignment submission, and grading. The system is built using the Layered Architecture Pattern, separating responsibilities into four distinct layers that includes Presentation, Business Logic, Data Access, and Database layer.

The prototype is fully deployed across three cloud platforms:

- a. Frontend: React with Vite which deployed on Netlify
- b. Backend: Spring Boot which deployed on Render
- c. Database Layer: Aiven MySQL cloud-managed relational database

1.2 Individual Role Summary

As the Frontend and Database Lead, my primary responsibilities throughout the project covers:

- a. Designing and developing the React frontend components including the Login page, Student Dashboard, and Instructor Console using Vite
- b. Implementing Axios HTTP client integration in api.js to connect the frontend to the Spring Boot backend via RESTful API calls
- c. Designing the database schema including all five entity tables: users, courses, assignments, contents, and submissions
- d. Developing the entity model classes (User.java, Course.java, Assignment.java, Content.java, Submission.java) and their relationships
- e. Deploying and configuring the React frontend on Netlify with continuous deployment from GitHub
- f. Contributing to the architecture report and system documentation

2. Weekly Progress

This section documents the weekly activities, individual contributions, challenges encountered, and action plans taken across all project phases.

2.1 Phase 1 — Project Proposal (Week 5)

Phase 1 Week 5 Project Proposal	
Tasks Undertaken	The project scope is discussed within the team to address traditional monolithic LMS design flaws, drafted the initial project proposal covering system objectives, and outlined the architectural approach alongside functional and non-functional requirements.
Individual Contributions	Contributed directly to defining the functional requirements for the four core modules and helped identify the frontend technology stack, choosing React with Vite for development and selecting the cloud hosting platform Netlify.
Challenges Faced	The team had difficulty deciding which frontend framework to use because we needed to balance the different skill levels within the team against the strict project timeline, which led to a long debate between using plain HTML/CSS/JavaScript versus a React-based Single Page Application.
Action Plan	The team agreed to use React with Vite after confirming it was beginner-friendly and easy to understand, and because Netlify natively supports static React builds with simple GitHub-based continuous deployment.

2.2 Phase 2 — Architecture Design (Week 7)

Phase 2 Week 7 Architecture Design	
Tasks Undertaken	The database schema was designed for all five entity tables, produced the entity relationship model for the database layer, and contributed to the Service and Component Identification section of the final report.
Individual Contributions	The full database schema was designed including the users, courses, assignments, contents, and submissions tables, and mapped out complex entity relationships like the many-to-one relationship between assignments and courses, as well as the many-to-many relationship between students and courses that auto-generates the course_enrollments junction table.
Challenges Faced	Deciding how to correctly model the student-course enrollment relationship in the database, because a simple foreign key approach would not support a scenario where many students need to enroll in many different courses simultaneously.
Action Plan	Researched JPA Many-to-Many relationship mapping and determined that annotating the enrolledStudents field in Course.java with @ManyToMany would instruct Hibernate to automatically generate a course_enrollments junction table to handle the many-to-many relationship correctly.

2.3 Project Observation 1 (Week 9)

Project Observation 1 Week 9	
Tasks Undertaken	The React frontend project was set up using Vite, developed the Login and Registration pages featuring form validation and role selection, configured Axios in api.js to connect to the Spring Boot backend, and presented the initial frontend progress to the lecturer.
Individual Contributions	A complete Login.jsx component was built with registration/login form logic, a role-based dropdown specifying STUDENT or INSTRUCTOR states, and error/success banner states, while also configuring api.js with the correct Render backend base URL and implementing the course card grid layout.
Challenges Faced	The frontend was successfully sending registration requests but the backend kept returning errors. The root cause was that the payload field name password did not match the backend model field passwordHash, and the role values Student and Instructor did not match the backend enum STUDENT and INSTRUCTOR.
Action Plan	The registration payload in Login.jsx is corrected to send passwordHash instead of password, all role values to uppercase STUDENT and INSTRUCTOR was updated to match the backend enum, and the login redirect logic to check role === 'INSTRUCTOR' is also updated.

2.4 Phase 3 — Prototype Progress (Week 11)

Phase 3 Week 11 Prototype Progress	
Tasks Undertaken	The frontend development was continued by implementing a complete Student Dashboard with a course catalog display and enrollment functionality, built a complete Instructor Console with a course creation form, and deployed the React frontend to Netlify via GitHub continuous deployment to test the live end-to-end system flow.
Individual Contributions	StudentDashboard.jsx is updated to fetch and display available courses using Axios GET /api/courses with custom course cards, implemented the enrollment button by calling POST /api/courses/{courseId}/enroll with the student's userId, and finalized the instructor course creation form with the correct backend field names.
Challenges Faced	After connecting the frontend to the live Render backend, the course cards were rendering completely blank even though the API was returning data, because the course field names used in the JSX (c.title, c.code, c.instructorName) did not match the actual field names returned by the backend payload (c.courseName, c.courseCode, c.instructor.name).
Action Plan	All the field references in StudentDashboard.jsx and InstructorDashboard.jsx are updated to use the correct backend payload field names (c.courseId, c.courseCode, c.courseName, and c.instructor?.name) and successfully fixed a maxWidth CSS typo that was breaking the layout.

2.5 Project Observation 2 (Week 12)

Project Observation 2 Week 12	
Tasks Undertaken	The Student Dashboard is improved to separate enrolled courses from available courses, tested all frontend API calls on the live deployed system to ensure they returned correct data, and resolved any remaining UI issues before the final freeze.
Individual Contributions	StudentDashboard.jsx is updated to call the new GET /api/courses/enrolled/{studentId} endpoint on page load to fetch enrolled courses separately, and implemented a tabbed interface featuring a "My Courses" tab with an Enrolled badge and an "Available Courses" tab that hides the enroll button for already-enrolled courses.
Challenges Faced	After a team member pushed changes to the GitHub repository for local testing, the api.js baseURL was accidentally changed to http://localhost:8080/api, causing the live Netlify site to auto-deploy the change and point to localhost instead of the Render backend, which completely broke the live application for all users.
Action Plan	The incorrect baseURL in api.js on GitHub is corrected by editing it directly in the GitHub web interface to restore the correct live Render URL (https://cloud-lms-backend.onrender.com/api), which allowed Netlify to auto-redeploy and restore the live site within 30 seconds.

2.6 Phase 4 — Project Report (Week 13)

Phase 4 Week 13 Project Report & Individual Logbook	
Tasks Undertaken	Contributed to writing the project report by taking responsibility for the Functional Requirements section, the Technology Stack section, and the presentation layer components in the Service and Component Identification section, while also compiling the individual logbook.
Individual Contributions	Wrote the Functional Requirements module descriptions for the Course Management and Content Delivery modules, documented the presentation layer tools including React, Vite, and Axios with their specific engineering roles, and detailed the final database schema containing all five entity tables and their exact relationships.
Challenges Faced	The functional requirements section was ensured to only described features that were actually implemented in the prototype, since some planned features such as file upload and profile editing were not completed.
Action Plan	The frontend code was reviewed to confirm which features were functional and adjusted the report to replace file upload with URL-based content upload, and profile editing was removed from the requirements section.

2.7 Phase 5 — Final Demonstration (Week 14)

Phase 5 Week 14 Final Demonstration	
Tasks Undertaken	The final system demonstration was prepared, the Netlify frontend was ensured live and correctly connected to the Render backend, and rehearsed the full user flow from registration to course enrollment and content access on the live deployed system.
Individual Contributions	The frontend user flows was demonstrated during the presentation including student registration, login, course catalog browsing, course enrollment, and the My Courses tab, and explained the React component structure , Axios API integration, and how the presentation layer communicates with the backend exclusively through RESTful API endpoints.
Challenges Faced	The live demo environment required all three cloud platforms to be active simultaneously, meaning any one of them being slow or unresponsive due to free-tier constraints could break the fluid demonstration flow in front of the lecturer.
Action Plan	Tested the full end-to-end flow on the live system the night before the demonstration to confirm everything was working, and kept the Netlify site open in a browser tab prior to the presentation to ensure the free-tier Render backend was pre-warmed and ready to respond instantly.

3. Generative AI Usage Record

All use of Generative AI tools during this project is transparently declared below. All AI-generated outputs were reviewed, verified, and adapted before being applied to any part of the project or report.

Total declared GenAI interactions: 6 entries across Weeks 5 to 13.

3.1 Entry 1 — Week 5

GenAI Entry 1 Week 5 Phase 1: Project Proposal	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	To get suggestions for the frontend technology stack suitable for a cloud-native uni project prototype
Prompt Used	"What frontend tools are beginner friendly for a university project using Spring Boot backend? We need to deploy to the cloud."
Output Summary	Recommended React with Vite as thhe frontend framework and Netlify as the hosting platform, explaining that Vite produces a static build that Netlify can serve, and that React integrates well with Spring Boot REST APIs via Axios.
How It Was Used	The recommendation confirms the team's choice of React with Vite and Netlify. The explanation was used to justify the technology choices in the proposal's technology stack section.

3.2 Entry 2 — Week 7

GenAI Entry 2 Week 7 Phase 2: Architecture Design	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	To understand how to model the many-to-many student-course enrollment relationship in JPA
Prompt Used	"How do I model a many to many relationship between students and courses in Spring Boot JPA so that enrolled students are stored correctly?"
Output Summary	Explained the <code>@ManyToMany</code> JPA annotation and how adding <code>enrolledStudents</code> as a List of User objects in <code>Course.java</code> would cause Hibernate to automatically create a <code>course_enrollments</code> junction table in MySQL.
How It Was Used	The <code>@ManyToMany</code> annotation was applied in <code>Course.java</code> to define the enrollment relationship. The auto-generated <code>course_enrollments</code> table was verified in the Aiven MySQL database after the first deployment.

3.3 Entry 3 — Week 9

GenAI Entry 3 Week 9 Project Observation 1	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	To fix the frontend registration and login failures after connecting to the live backend
Prompt Used	"My registration form keeps failing. The backend expects passwordHash but I am sending password. Also the role needs to be STUDENT not Student. Give me the corrected Login.jsx"
Output Summary	Produced a corrected Login.jsx with the registration payload field renamed from password to passwordHash, role values updated to uppercase STUDENT and INSTRUCTOR throughout the form and redirect logic, and the onLoginSuccess prop wired correctly to update the App state after login.
How It Was Used	The Login.jsx file in the project is replaced with the corrected version, pushed to GitHub, and verified that registration and login worked correctly on the live Netlify and Render deployment.

3.4 Entry 4 — Week 11

GenAI Entry 4 Week 11 Phase 3: Prototype Progress	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	To fix the blank course cards on the Student Dashboard after connecting to the live backend
Prompt Used	"My course cards are showing blank. The backend returns courseId, courseName, courseCode and instructor object but my JSX uses c.id, c.title, c.code. Give me the corrected StudentDashboard and InstructorDashboard"
Output Summary	It produces a corrected StudentDashboard.jsx and InstructorDashboard.jsx with all field references updated to match the backend response (c.courseId, c.courseName, c.courseCode, c.instructor?.name). Also fixed the maxWIdth typo and corrected the course creation payload to send courseName, courseCode, and instructor.userId.
How It Was Used	Both dashboard files are replaced with the corrected versions, pushed to GitHub, and verified that course cards rendered correctly with all data visible on the live deployed Netlify site.

3.5 Entry 5 — Week 12

GenAI Entry 5 Week 12 Project Observation 2	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	In order to improve the student dashboard to show enrolled courses separately from available courses
Prompt Used	"Improve my code where once enrolled into a course it actually stores the course data under that user"
Output Summary	It produced an updated StudentDashboard.jsx with a tabbed interface separating My Courses (enrolled) from Available Courses (not yet enrolled), fetching enrolled courses via a new GET /api/courses/enrolled/{studentId} endpoint, and hiding the enroll button for already-enrolled courses.
How It Was Used	StudentDashboard.jsx is replaced with the improved version, pushed to GitHub, and confirmed on the live site that enrolled courses appeared under the My Courses tab after enrollment and disappeared from the Available Courses tab.

3.6 Entry 6 — Week 13

GenAI Entry 6 Week 13 Phase 4: Project Report	
Tool Used	Claude (Anthropic) — claude.ai
Purpose	To update the Functional Requirements and Technology Stack sections of the report to reflect the actual implemented system
Prompt Used	"I will give you parts of my report, only change it based on how we developed the project and the right tools. Mostly keep my sentences."
Output Summary	It updates the Functional Requirements section to remove unimplemented features such as file upload and profile editing, and updates the Technology Stack section to correctly name React, Vite, Axios, and Netlify with their specific roles in the presentation layer.
How It Was Used	All AI-updated paragraphs are reviewed against the actual frontend codebase before including them in the final report.

4. Academic Integrity Declaration

I hereby declare that all work documented in this individual logbook is a true and accurate record of my contributions to the Cloud-Native Learning Management System group project. I acknowledge and accept the following:

All tasks, contributions, and activities recorded in Section 2 represent my genuine individual work and involvement in the project.

All use of Generative AI tools has been fully and transparently declared in Section 3, including the tool used, the purpose, the prompt submitted, the output received, and how the output was applied.

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I confirm that the contributions attributed to me in the Team Contributions section of the group report are accurate and reflect my actual role in the project.

Full Name	Mohammad Adam Basheer B. Mohd Azrul
Matrix No	S75710
Group Number	Group 4
Academic Role	Frontend & Database Lead

INDIVIDUAL PROJECT LOGBOOK

Project Title: Cloud-Native Learning Management System (LMS) with Layered Architecture

Student Name: Ammar Haziq Bin Saiful Bahri

Academic Role: Full Stack & Architecture Lead

Group Number: Group 4

1. Team Information

Team Members	Role
Eashenraj A/L Nagarajan	Backend & Deployment Lead
Mohammad Adam Basheer B. Mohd Azrul	Frontend & Database Lead
Ammar Haziq B. Saiful Bahri	Full Stack & Architecture Lead

1.1 Project Overview

The Cloud-Native Learning Management System (LMS) is a platform developed to manage educational activities for Students and Instructors which includes course creation, student enrolment, assignment submission, and grading. The system is built using the Layered Architecture Pattern, separating responsibilities into four distinct layers which are Presentation, Business Logic, Data Access, and Database.

The prototype is fully deployed across three cloud platforms:

- Frontend (Presentation Layer): React with Vite, deployed on Netlify
- Backend (Business Logic + Data Access Layer): Spring Boot, deployed on Render
- Database Layer: Aiven MySQL cloud-managed relational database

1.2 Individual Role Summary

As the Full Stack and Architecture Lead, my primary responsibilities throughout the project covered:

- Lead the architecture design of the Cloud LMS system ensuring proper separation between each layer (presentation, business logic, database)
- Developing frontend dashboard views for both the Student and Instructor roles
- Integrating and testing the full stack end-to-end of the Cloud LMS flow connecting React, Spring Boot, and Aiven MySQL
- Leading the architecture report sections and ensuring all documentation accurately reflected the implemented Cloud Learning Management System.

2. Weekly Progress

This section documents the weekly activities, individual contributions, challenges encountered, and action plans undertaken across all project phases.

2.1 Phase 1 — Project Proposal (Week 5)

Phase 1 Week 5 Project Proposal	
Tasks	Held a team discussion to discuss about the scope of project and architecture style choice. Decided to go with layered architecture style for the implementation of Cloud Learning Management System. Drafted a report with teammates for the architecture style and objectives in the project proposal.
Individual Contributions	Wrote the architecture style justification section explaining why the Layered Architecture Pattern was chosen. Defined the four-layer structure which are presentation, business logic, data access and database. Drafted the non-functional requirements section which covers performance, scalability, security, availability, and maintainability.
Challenges Faced	Deciding how to clearly justify the implementation of layered architecture.
Action Plan	Researched important information of layered architecture and separation of concerns.

2.2 Phase 2 — Architecture Design (Week 7)

Phase 2 Week 7 Architecture Design	
Tasks	Led the system architecture design. Produced the final UML deployment diagram in VisualParadigm. Defined all service components and DAO components across the four layers. Wrote the Service and Component Identification section of the report.
Individual Contributions	Designed and produced the UML deployment diagram using VisualParadigm. Identify and apply the Business Logic Layer and Data Access Layer correctly inside Render backend container. Wrote the full Service and Component Identification section describing all four service components and four DAO components with their responsibilities.
Challenges Faced	Determining the correct UML to use for each node and component in the deployment diagram, and correctly representing the physical deployment across Netlify, Render, and Aiven.
Action Plan	Researched UML 2.0 deployment diagram conventions and confirmed the correct stereotypes for each node type. Used the AI-generated diagram reference as a structural guide and recreated the final version manually in VisualParadigm to ensure UML compliance.

2.3 Project Observation 1 (Week 9)

Project Observation 1 Week 9	
Tasks	Implemented all five Spring Boot REST API controllers and all five Spring Data JPA repository interfaces. Ensured all endpoints were correctly mapped and returning appropriate HTTP responses. Began integration testing between the frontend and backend. Presented architecture and backend structure to the lecturer.
Individual Contributions	Implemented UserController with /api/users/register and /api/users/login endpoints. Implemented CourseController with course creation, catalog retrieval, and enrollment endpoints. Implemented AssignmentController, ContentController, and SubmissionController with their respective CRUD and business endpoints. Implemented all five JPA repository interfaces including custom JPQL queries such as findByMatricNo and findByCourseCourseId.
Challenges Faced	The CourseController enrollment endpoint was receiving the studentId from the request body but the service layer was throwing a null pointer exception. The issue was that the payload key name in the frontend did not match what the controller was reading from the Map.
Action Plan	Updated the CourseController enroll method to read studentId from the Map payload using payload.get("studentId") and cast it to Integer. Verified the fix using Postman by sending a test POST request with the correct JSON body before connecting the frontend.

2.4 Phase 3 — Prototype Progress (Week 11)

Phase 3 Week 11 Prototype Progress	
Task	Completed the integration. Implemented SubmissionService with automatic deadline-based status tracking. Developed the Instructor Console dashboard views for viewing submitted assignments and providing grades and feedback. Conducted end-to-end integration testing of all four modules on the live deployed system.
Individual Contributions	Implemented SubmissionService.submitAssignment() with automatic LocalDateTime comparison against the assignment due date to set status to SUBMITTED or LATE. Implemented SubmissionService.gradeSubmission() to update grade and feedback fields. Developed the grading view in the Instructor Console showing student submissions with grade and feedback input fields. Conducted Postman testing across all 12 API endpoints to verify correct responses before demo.
Challenges Faced	The grading endpoint was updating the submission record in the database but the student dashboard was not reflecting the updated grade. The issue was that the student dashboard was only fetching submissions on initial page load and was not refreshing after a grade update.
Action Plan	Added a refresh mechanism to the student submissions view that re-fetches submission data from GET /api/submissions/student/{studentId} after a grade update is detected, ensuring the displayed grade and feedback were always current.

2.5 Project Observation 2 (Week 12)

Project Observation 2 Week 12	
Task	Conducted the system integration across all layers. Tested the flow of the system from frontend (React) through Spring Boot controllers, service layer, repository layer and database (Aiven MySQL)
Individual Contributions	Added a new GET /api/courses/enrolled/{studentId} endpoint in CourseController and implemented the corresponding getEnrolledCourses method in CourseService and the findEnrolledCoursesByStudentId JPQL query in CourseRepository. Also added duplicate enrollment prevention in CourseService.enrollStudent() using a stream filter to check if the student was already present in the enrolledStudents list before saving.
Challenges Faced	During integration testing, the WebConfig.java CORS configuration allowed the frontend to reach the backend in local development but unable to reach and obtain request from the live Netlify URL on the deployed system.
Action Plan	Updated the allowedOrigins value in WebConfig.java to exactly match the live Netlify URL including the correct subdomain. Redeployed the backend of Cloud LMS to Render via GitHub push.

2.6 Phase 4 — Project Report (Week 13)

Phase 4 Week 13 Project Report & Individual Logbook	
Tasks Undertaken	Led the compilation of the project report. Wrote the Architecture Style, Service and Component Identification, and Architecture Evaluation sections. Reviewed and aligned all report sections across the team to ensure consistency with the implemented system.
Individual Contributions	Wrote the Architecture Style section explaining the closed layered approach and each layer's responsibility with the tools that are used. Wrote the Architecture Evaluation section verifying the four-layer separation of concerns pattern and NFR validation. Reviewed all team members' report sections.
Challenges Faced	Coordinating across three team members to ensure all sections of the report used consistent terminology, tool names, and class names.
Action Plan	Held a team review session where each member cross-checked their assigned parts of the Cloud LMS system.

2.7 Phase 5 — Final Demonstration (Week 14)

Phase 5 Week 14 Final Demonstration	
Tasks Undertaken	Led the preparation and delivery of the final system demonstration. Prepared the full-stack walkthrough covering the architecture design, backend code structure, REST API endpoints, and live system flows. Ensured all cloud services were active and responsive before the demonstration.
Individual Contributions	Presented the system architecture and explained the Layered Architecture Pattern and separation of concerns to the lecturer. Walked through the backend controller, service, and repository package structure explaining each layer's responsibility.
Challenges Faced	Explaining the relationship between the business logic layer and the data access layer clearly.
Action Plan	Used the information that are gathered throughout the project which includes the tools used, UML diagram and also requirements for the Cloud Learning Management System.

3. Generative AI Usage Record

During the developing phase of the project, Generative AI tools are used to aid the development of our Cloud Learning Management System. All AI-generated prompt input and output are recorded in the tables below

3.1 Entry 1 — Week 5

GenAI Entry 1 Week 5 Phase 1: Project Proposal	
Tool Used	Claude
Purpose	To review the project report flow
Prompt Used	"Read my report and tell me it is structured and organized in a good flow"
Output Summary	Claude AI identified that Postman was incorrectly described as a REST API and a few repetitions in the problem description. It suggested to restructure the paragraph separate each core problems.
How It Was Used	Restructure the problem description section and correct the Postman description. Recheck with the other team members before submission.

3.2 Entry 2 — Week 7

GenAI Entry 2 Week 7 Phase 2: Architecture Design	
Tool Used	Claude
Purpose	Generate the important details of UML deployment diagram for architectural design reference.
Prompt Used	"Tell the important details of UML deployment diagram for a system that is using React, Spring Boot, and MySQL on Netlify, Render, and Aiven"
Output Summary	Listed UML deployment diagram information with stereotypes including the component, artifact, database, and table. Apply the Business Logic Layer and DAL inside the backend container with communication links labelled with protocols.
How It Was Used	Used as a structural reference for UML diagram. The final deployment diagram was created in Visual Paradigm based on this reference to make sure it tally with the system.

3.3 Entry 3 — Week 7

GenAI Entry 3 Week 7 Phase 2: Architecture Design	
Tool Used	Claude
Purpose	Identify and structure all service and component descriptions for the report section
Prompt Used	"Since you have the whole idea of the system, what are all the service components? I need to do a report part for Service and Component Identification"
Output Summary	Produced a structured breakdown of four modules with service components in the Business Logic Layer and DAO components in the Data Access Layer, with a summary table mapping each component to its layer and responsibility.
How It Was Used	Used as the foundation for Section 4.2 Service and Component Identification. All components were verified against the actual implemented classes and the section was rewritten in the team's own words before being included in the report.

3.4 Entry 4 — Week 9

GenAI Entry 4 Week 9 Project Observation 1	
Tool Used	Claude
Purpose	Understand the function of each backend package to increase my understanding.
Prompt Used	"What is the function of all backend packages model, repo and all and how does it actually work"
Output Summary	Produced a detailed explanation of all four backend packages: model, repository, service, and controller. Included a full request-response flow example using course enrolment.
How It Was Used	Used to carefully understand the implementation of backend. Provide a better understanding to myself and the other team members.

3.5 Entry 5 — Week 11

GenAI Entry 5 Week 11 Phase 3: Prototype Progress	
Tool Used	Claude
Purpose	Perform a full code audit of all backend and frontend files to identify any remaining bugs before the final demo
Prompt Used	"Recheck everything in details and coding lines to ensure it is bug free"
Output Summary	Identified few issues across the full codebase such as both dashboard files using wrong field names from the backend, InstructorDashboard course creation sending wrong field names, and the missing email null guard in UserService.
How It Was Used	Fixed the bugs that are identified and continue to push it into the git hub repository.

3.6 Entry 6 — Week 13

GenAI Entry 6 Week 13 Phase 4: Project Report	
Tool Used	Claude
Purpose	Tolls suggestion to validate the NFR
Prompt Used	"Based on my system I want you to suggest me a few tools to help me validate my NFR for Cloud LMS"
Output Summary	Suggested to use few applications such as UptimeRobot to monitor system uptime (Availability) and AutoCannon for response time (Performance) .
How It Was Used	Followed the instruction from cloud to run the application and recorded the result to display it and validate the system in the final report.

4. Academic Integrity Declaration

I hereby declare that all work documented in this individual logbook is a true and accurate record of my contributions to the Cloud-Native Learning Management System group project. I acknowledge and accept the following:

- All tasks, contributions, and activities recorded in Section 2 represent my genuine individual work and involvement in the project.
- All use of Generative AI tools has been fully and transparently declared in Section 3, including the tool used, the purpose, the prompt submitted, the output received, and how the output was applied.
- No AI-generated content was submitted directly without review. All outputs were critically evaluated and adapted before being applied to any deliverable.
- I understand that any undeclared use of Generative AI tools or misrepresentation of individual contributions constitutes a breach of the academic integrity policy.

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